Computer Project 1

Write a code for solving the equation

$$u_t + a(x,t)u_x = 0, \quad 0 \le x \le 1,\ 0 \le t \le 1, \tag{1}$$

where $a(x,t) \ge 0$, with the initial condition

$$u(x,0) = u_0(x) \tag{2}$$

and the boundary condition

$$u(0,t) = 0, \tag{3}$$

using the upwind method. Take

$$a(x,t) = \frac{1+x^2}{1+2xt+2x^2+x^4}$$

and

$$u_0(x) = \begin{cases} 1 & \text{if } 0.2 \le x \le 0.4, \\ 0 & \text{otherwise.} \end{cases}$$

The exact solution of the problem is

$$u(x,t) = u_0(x^*), \quad x^* = x - \frac{t}{1+x^2}.$$

- Take $\Delta t = \Delta x$. Check that this choice satisfies the CFL condition.
- Run the code with $\Delta x = 0.02, 0.01$. Plot the computed and true solutions at $t = 0.5$ and $t = 1$.
- Modify your code to implement the Lax-Wendroff scheme. Run the modified code with $\Delta x = 0.02, 0.01$. Plot the computed and true solutions at $t = 0.5$ and $t = 1$.

- Now take the initial condition

$$u_0(x) = e^{-10(4x-1)^2}.$$

  Redo the above runs and plots with both methods.

- Estimate the convergence of both methods in each of the two cases. Assume relation for the error of the form

$$E(\Delta x) = \max_j |U(x_j, 1) - u(x_j, 1)| \leq C\Delta x^p \tag{4}$$

  Run the code with $\Delta x = 0.1, 0.05, 0.025, 0.0125$. Compute $E(\Delta x)$ and use the results to estimate C and p in (4).

Submit the following

- A listing of your code
- Plots for the two methods with the two initial conditions, and the two time steps - total of eight plots
- A discussion on the quality of the solutions in the plots. For each method discuss if refining the grid improves the solution. Also compare the solutions from the two methods.
- The computed $E(\Delta x)$ and the estimated C and p. Do the numerical convergence rates correspond to the theoretical ones? If not, why not ?